

CSC - 321

DESIGN AND ANALYSIS OF ALGORITHMS



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Longest Common Subsequence

- ▶ In biological applications, we often want to compare the DNA of two (or more) different organisms.
- ▶ A strand of DNA consists of a string of molecules called bases, where the possible bases are adenine, guanine, cytosine, and thymine (A, G, C, T).
- ▶ Comparison of two DNA strings
 - ❖ $S_1 = \text{ACCGGTCGAGTGCGCGGAAGCCGGCCGAA}$
 - ❖ $S_2 = \text{GTCGTTCGGAATGCCGTTGCTCTGTAAA}$
 - ❖ $S_3 = \text{GTCGTTCGGAAGCCGGCCGAA}$
- Similarity can be defined in different ways:
 - ❖ Two DNA strands are similar if one is a substring of the other.
 - ❖ Two strands are similar if the number of changes needed to turn one into the other is small.
 - ❖ There is a third strand S_3 in which the bases in S_3 appear in each of S_1 and S_2 ; these bases must appear in the same order, but not necessarily consecutively. The longer the strand S_3 we can find, the more similar S_1 and S_2 are. (we focus on this)

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Longest Common Subsequence

Introduction

- ▶ A subsequence is a sequence that appears in the same relative order, **but not necessarily contiguous**.
- ▶ Given two sequences X and Y , we say that a sequence Z is a common subsequence of X and Y if Z is a subsequence of both X and Y .
- ▶ E.g., if $X = \langle \underline{A}, \underline{B}, C, B, \underline{D}, A, B \rangle$ and $Y = \langle \underline{B}, \underline{D}, C, A, B, \underline{A} \rangle$ then $\langle B, C, A \rangle$ is a subsequence.
- ▶ Use dynamic programming technique to find **the longest common subsequence (LCS)**.

Longest Common Subsequence

- ▶ We need to generate table $c(1..m, 1..n)$
where m = length of string $S1$ and n = length of string $S2$.

To generate table $c[i][j]$ use following steps:

Step-1: Make $c[i][0] = 0$ and $c[0][j] = 0$

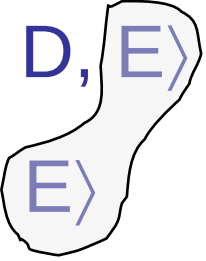
Step-2: if $x_i = y_j$ then $c[i, j] \leftarrow c[i - 1, j - 1] + 1$

Step-3: else $c[i, j] \leftarrow \max(c[i - 1, j], c[i, j - 1])$

A Recursive Solution

Case 1: $x_i = y_j$

e.g.: $X_i = \langle A, B, D, E \rangle$
 $Y_j = \langle Z, B, E \rangle$



$$c[i, j] = c[i - 1, j - 1] + 1$$

- Append $x_i = y_j$ to the LCS of X_{i-1} and Y_{j-1}
- Must find a LCS of X_{i-1} and $Y_{j-1} \Rightarrow$ optimal solution to a problem includes optimal solutions to subproblems

A Recursive Solution

Case 2: $x_i \neq y_j$

e.g.: $X_i = \langle A, B, D, G \rangle$

$Y_j = \langle Z, B, D \rangle$

$$c[i, j] = \max \{ c[i - 1, j], c[i, j - 1] \}$$

- Must solve two problems
 - find a LCS of X_{i-1} and Y_j : $X_{i-1} = \langle A, B, D \rangle$ and $Y_j = \langle Z, B, D \rangle$
 - find a LCS of X_i and Y_{j-1} : $X_i = \langle A, B, D, G \rangle$ and $Y_j = \langle Z, B \rangle$
- Optimal solution to a problem includes optimal solutions to subproblems

3. Computing the Length of the LCS

$$c[i, j] = \begin{cases} 0 & \text{if } i = 0 \text{ or } j = 0 \\ c[i-1, j-1] + 1 & \text{if } x_i = y_j \\ \max(c[i, j-1], c[i-1, j]) & \text{if } x_i \neq y_j \end{cases}$$

		0	1	2	n	
		$y_j:$	y_1	y_2	y_n	
0	x_i	0	0	0	0	0
1	x_1	0	→			
2	x_2	0	→			
		0				
		0				
m	x_m	0	→			

j

first
second
i

Additional Information

$$c[i, j] = \begin{cases} 0 & \text{if } i, j = 0 \\ c[i-1, j-1] + 1 & \text{if } x_i = y_j \\ \max(c[i, j-1], c[i-1, j]) & \text{if } x_i \neq y_j \end{cases}$$

b & c:

	0	1	2	3	n
y_j :	A	C	D	F	
0 x_i	0	0	0	0	0
1 A	0				
2 B	0			$c[i-1, j]$	
3 C	0		$c[i, j-1]$		
	0				
m D	0				

j

i

A matrix $b[i, j]$:

- For a subproblem $[i, j]$ it tells us what choice was made to obtain the optimal value
- If $x_i = y_j$
 $b[i, j] = \nwarrow$
- Else, if
 $c[i-1, j] \geq c[i, j-1]$
 $b[i, j] = \uparrow$
 else
 $b[i, j] = \leftarrow$

LCS-LENGTH(X, Y, m, n)

```
1. for i ← 1 to m
2.   do c[i, 0] ← 0
3. for j ← 0 to n
4.   do c[0, j] ← 0
5. for i ← 1 to m
6.   do for j ← 1 to n
7.     do if  $x_i = y_j$ 
8.       then  $c[i, j] \leftarrow c[i - 1, j - 1] + 1$ 
9.          $b[i, j] \leftarrow \nwarrow$ 
10.    else if  $c[i - 1, j] \geq c[i, j - 1]$ 
11.      then  $c[i, j] \leftarrow c[i - 1, j]$ 
12.         $b[i, j] \leftarrow \uparrow$ 
13.    else  $c[i, j] \leftarrow c[i, j - 1]$ 
14.       $b[i, j] \leftarrow \leftarrow$ 
15. return c and b
```

The length of the LCS if one of the sequences is empty is zero

Case 1: $x_i = y_j$

Case 2: $x_i \neq y_j$

Running time: $\Theta(mn)$

Example

$$X = \langle A, B, C, B, D, A \rangle$$

$$Y = \langle B, D, C, A, B, A \rangle$$

$$c[i, j] = \begin{cases} 0 & \text{if } i = 0 \text{ or } j = 0 \\ c[i-1, j-1] + 1 & \text{if } x_i = y_j \\ \max(c[i, j-1], c[i-1, j]) & \text{if } x_i \neq y_j \end{cases}$$

If $x_i = y_j$

$b[i, j] = "$ ↖ "

Else if

$c[i-1, j] \geq c[i, j-1]$

$b[i, j] = "$ ↑ "

else

$b[i, j] = "$ ← "

		0	1	2	3	4	5	6
		Y_j	B	D	C	A	B	A
0	x_i	0	0	0	0	0	0	0
1	A	0	$\uparrow$ 0	$\uparrow$ 0	$\uparrow$ 0	$\swarrow$ 1	$\leftarrow$ 1	$\swarrow$ 1
2	B	0	$\swarrow$ 1	$\leftarrow$ 1	$\leftarrow$ 1	$\uparrow$ 1	$\swarrow$ 2	$\leftarrow$ 2
3	C	0	$\uparrow$ 1	$\uparrow$ 1	$\swarrow$ 2	$\leftarrow$ 2	$\uparrow$ 2	$\uparrow$ 2
4	B	0	$\swarrow$ 1	$\uparrow$ 1	$\uparrow$ 2	$\uparrow$ 2	$\swarrow$ 3	$\leftarrow$ 3
5	D	0	$\uparrow$ 1	$\swarrow$ 2	$\uparrow$ 2	$\uparrow$ 2	$\uparrow$ 3	$\uparrow$ 3
6	A	0	$\uparrow$ 1	$\uparrow$ 2	$\uparrow$ 2	$\swarrow$ 3	$\uparrow$ 3	$\swarrow$ 4
7	B	0	$\swarrow$ 1	$\uparrow$ 2	$\uparrow$ 2	$\uparrow$ 3	$\swarrow$ 4	$\uparrow$ 4

4. Constructing a LCS

- Start at $b[m, n]$ and follow the arrows
- When we encounter a “ $\nwarrow$ ” in $b[i, j] \Rightarrow x_i = y_j$ is an element of the LCS

		0	1	2	3	4	5	6
	y_j	B	D	C	A	B	A	
0	x_i	0	0	0	0	0	0	
1	A	0	↑ 0	↑ 0	↑ 0	↖ 1	← 1	↖ 1
2	B	0	↖ 1	← 1	← 1	↑ 1	↖ 2	← 2
3	C	0	↑ 1	↑ 1	↖ 2	← 2	↑ 2	↑ 2
4	B	0	↖ 1	↑ 1	↑ 2	↑ 2	↖ 3	← 3
5	D	0	↑ 1	↖ 2	↑ 2	↑ 2	↑ 3	↑ 3
6	A	0	↑ 1	↑ 2	↑ 2	↖ 3	↑ 3	↖ 4
7	B	0	↖ 1	↑ 2	↑ 2	↑ 3	↖ 4	↑ 4

Example 2

$X = \langle A, B, C, B, D, A, B \rangle$ and $Y = \langle B, D, C, A, B, A \rangle$ and $LCS = \langle B, D, A, B \rangle$

	y_j	A	B	C	B	D	A	B
x_i	0	0	0	0	0	0	0	0
B	0	↑ 0	↖ 1	← 1	↖ 1	← 1	← 1	↖ 1
D	0	↑ 0	↑ 1	↑ 1	↑ 1	↖ 2	← 2	← 2
C	0	↑ 0	↑ 1	↖ 2	← 2	↑ 2	↑ 2	↑ 2
A	0	↖ 1	↑ 1	↑ 2	↑ 2	↑ 2	↖ 3	← 3
B	0	↑ 1	↖ 2	↑ 2	↖ 3	← 3	↑ 3	↖ 4
A	0	↖ 1	↑ 2	↑ 2	↑ 3	↑ 3	↖ 4	↑ 4

Example 2

		C	B	D	A
	0	0	0	0	0
A	0	0	0	0	1
C	0	1	1	1	1
A	0	1	1	1	2
D	0	1	1	2	2
B	0	1	2	2	2

